

MOHAMMED ALDKHILY

COMPUTER SCIENCE

Riyadh | +966 54 315 6466 | mohammed.aldkhily@gmail.com | imdk-cs.github.io/Portfolio

SUMMARY

Senior Computer Science student specializing in Artificial Intelligence and Machine Learning, with a growing focus on AI agent systems and modern AI patterns such as RAG, agentic workflows, and MCP. I build practical AI-driven applications and complement my AI work with full-stack development skills to deliver complete, production-ready solutions.

EDUCATION

Qassim University | Saudi Arabia

2021 – 2026

Bachelor's Degree in Computer Science

PROJECTS

Graduation Project, AI-Based Pre-consultation Medical Diagnosis System

2026

- Developed a bilingual (Arabic/English) web application using Next.js and OpenAI API to automate patient data collection.
- Implemented NLP to analyze free-text symptom descriptions and generate structured medical reports for doctors.
- Utilized Drizzle ORM and Neon DB for efficient data management and secure storage.

Personal Project, AI Smart Home Assistant | n8n + MCP + Tuya IoT

2026

- Built an AI-powered Telegram assistant that controls real-world IoT devices through natural language commands in Arabic/English.
- Implemented an MCP Server following Anthropic's latest agent integration standard. Integrated Tuya Cloud API for device control lighting, appliances and Gemini for natural language understanding.
- Designed the agent architecture with LLM reasoning, tool calling, and persistent memory achieved ~4 seconds average response time.

Personal Project, SmoothFlow | Full-Stack E-commerce Platform

2025

- Built a full-stack PC store platform using Next.js 15, TypeScript, and Prisma ORM with PostgreSQL.
- Applied rate limiting and security headers to protect against common web vulnerabilities.
- Designed a responsive modern UI with Tailwind CSS; achieved page load times under 2 seconds for 98% of queries.
- Deployed on Vercel with an automated CI/CD pipeline.

CERTIFICATIONS

- Artificial Intelligence Concepts and Advanced Applications – SDAIA
- Fundamentals of Artificial Intelligence – SDAIA
- Advanced AI and Machine Learning Techniques and Capstone – Microsoft
- Foundations of AI and Machine Learning – Microsoft
- RAG and Agentic AI Capstone Project – IBM
- Build AI Agents using MCP – IBM
- Agentic AI with LangGraph, CrewAI, AutoGen and BeeAI – IBM
- Fundamentals of Building AI Agents – IBM
- Build Multimodal Generative AI Applications – IBM
- Data Analysis – MISK

SKILLS

- **Programming Languages:** TypeScript, JavaScript, Python, C++, SQL
- **Frameworks & Libraries:** Next.js, React, Node.js, Tailwind CSS
- **Databases:** PostgreSQL, Prisma ORM, Drizzle ORM, Neon DB
- **AI & Automation:** OpenAI API, Google Gemini, MCP, n8n, NLP fundamentals
- **APIs & Integration:** RESTful APIs
- **Tools & Platforms:** Git, GitHub, Vercel, Postman
- **Other:** UI/UX Design, Microsoft Office

LANGUAGES

Arabic: Native **English:** Fluent